# BIOCHEMISTRY CUME
## September 24, 2004

1. (40 points) Many biochemical processes involve the interaction between two molecular species.

$$A + B \Leftrightarrow AB \qquad K_{association} = \text{equilibrium constant for reaction as written.}$$

Defining Y as the fraction of A molecules with B bound [Y = (AB)/(Atotal)], show that Y is related to $K_{assoc}$, the equilibrium constant for the above reaction by the following equation under conditions where (A) is kept constant and (B) is varied.

$$Y = \frac{K_{assoc}\,(B)}{1 + K_{assoc}\,(B)}$$

2. (30 points) In many biochemical systems, the equilibrium is often described in terms of $K_{dissociation}$, the equilibrium constant for the reverse of the reaction shown in #1. Starting with the equation you derived in #1 and using the relationship between $K_{assoc}$ and $K_{dissoc}$, show that:

$$Y = \frac{(B)}{K_{dissoc} + (B)}$$

3. (30 points) The dissociation of a weak acid

$$HA \Leftrightarrow A + H^+$$

is described by the Henderson-Hasselbach equation

$$pH = pKa + \log \frac{(A)}{(HA)}$$

where Ka is the equilbrium constant for the dissociation of HA.

Defining Y as the fraction of weak acid which has dissociated, show that:

$$Y = \frac{10^{pH-pKa}}{1 + 10^{pH-pKa}}$$